

MAIE5102 Assignment 1

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Question 1: Computation Graph and Gradients

Given the function:

$$f(x_0, x_1, x_2, x_3) = e^{x_3}(x_0 + 3x_1)x_2^2$$

with:

$$x_0 = 1, \quad x_1 = -1, \quad x_2 = 2, \quad x_3 = 16$$

Solution

Gradients

Let:

$$a = x_0 + 3x_1$$

$$b = x_2^2$$

$$c = e^{x_3}$$

$$f = c \cdot a \cdot b$$

Forward Pass:

$$a = 1 + 3(-1) = -2$$

$$b = 2^2 = 4$$

$$c = e^{16}$$

$$f = e^{16} \cdot (-2) \cdot 4 = -8e^{16}$$

Backward Pass:

$$\frac{\partial f}{\partial c} = ab = (-2)(4) = -8$$

$$\frac{\partial f}{\partial a} = cb = e^{16} \cdot 4 = 4e^{16}$$

$$\frac{\partial f}{\partial b} = ca = e^{16}(-2) = -2e^{16}$$

Now propagate to the inputs.

$$\frac{\partial f}{\partial x_3} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial x_3} = (-8)(e^{16}) = -8e^{16}$$

$$\frac{\partial f}{\partial x_2} = \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial x_2} = (-2e^{16})(2x_2) = (-2e^{16})(4) = -8e^{16}$$

$$\frac{\partial f}{\partial x_0} = \frac{\partial f}{\partial a} \cdot 1 = 4e^{16}$$

$$\frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial a} \cdot 3 = 12e^{16}$$

Final Results:

$$f = -8e^{16}$$

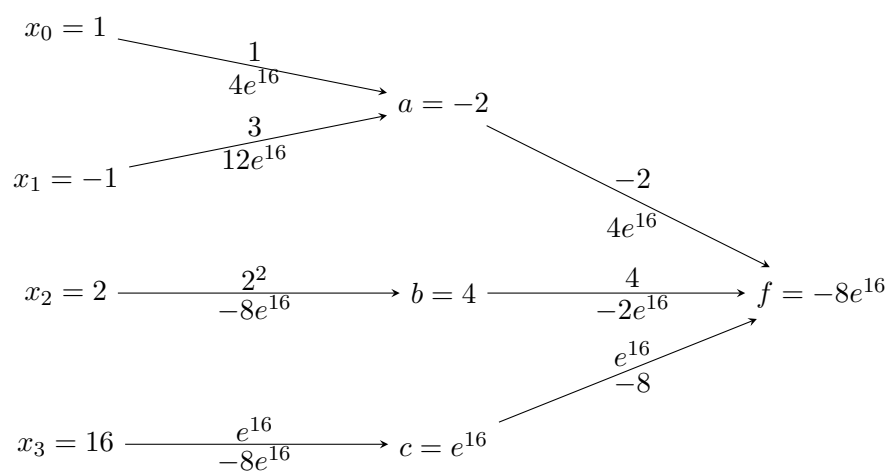
$$\frac{\partial f}{\partial x_0} = 4e^{16}$$

$$\frac{\partial f}{\partial x_1} = 12e^{16}$$

$$\frac{\partial f}{\partial x_2} = -8e^{16}$$

$$\frac{\partial f}{\partial x_3} = -8e^{16}$$

Computation Graph



Question 2: Observations on Hidden Units

Observation:

In this experiment, we trained one-hidden-layer MLPs with different numbers of hidden units (2, 5, 10, 50, 100, 500, 1000, and 5000) on the 2-cluster dataset and visualized their decision boundaries.

When the number of hidden units is small (e.g., 2), the decision boundary is nearly linear and lacks sufficient flexibility to capture the structure of the data. As a result, the model underfits and misclassifies some samples.

As the number of hidden units increases (e.g., 5 and 10), the decision boundary becomes more flexible and begins to better approximate the separation between the two clusters. The classification performance improves accordingly.

When the hidden units reach moderate values (e.g., 50 and 100), the model is expressive enough to separate the two clusters effectively. The decision boundary becomes smooth and well-aligned with the data distribution, and the testing accuracy stabilizes at a high level.

For very large numbers of hidden units (e.g., 500, 1000, and 5000), the decision boundary does not change significantly compared to the moderate-sized models. Although the model capacity increases substantially, the improvement in testing accuracy is marginal. This indicates diminishing returns when increasing model complexity beyond a certain point.

Overall, the experiment demonstrates that increasing the number of hidden units enhances model capacity and improves performance up to a certain level. However, excessively large models provide limited additional benefit while increasing computational cost.